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Practice : 1

In [1]:

```python
inputs = [1, 2, 3]
Wx, Wh, b = 0.5, 0.5, 0
h = 0
hidden_states = []

for x in inputs:
    h = (Wx * x) + (Wh * h) + b
    hidden_states.append(h)

print(f"Hidden states = {hidden_states}")

# Task 2: SimpleRNN model
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import SimpleRNN, Dense

model = Sequential([
    SimpleRNN(4, input_shape=(3, 1)),
    Dense(1)
])

model.compile(optimizer='adam', loss='mse')

print("Model created successfully with:")
print("  SimpleRNN layer (4 units)")
print("  Dense output layer (1 unit)")
model.summary()
```

```
Hidden states = [0.5, 1.25, 2.125]
Model created successfully with:
  SimpleRNN layer (4 units)
  Dense output layer (1 unit)
```

```
c:\Users\HP\AppData\Local\Programs\Python\Python312\Lib\site-packages\keras\src\l
ayers\rnn\rnn.py:199: UserWarning: Do not pass an `input_shape`/`input_dim` argum
ent to a layer. When using Sequential models, prefer using an `Input(shape)` obje
ct as the first layer in the model instead.
  super().__init__(**kwargs)
```

Model: "sequential"

Layer (type)	Output Shape	Param #
simple_rnn (SimpleRNN)	(None, 4)	24
dense (Dense)	(None, 1)	5

Total params: 29 (116.00 B)

Trainable params: 29 (116.00 B)

Non-trainable params: 0 (0.00 B)